Research paper

Advances in Coated Fertilizers and Their Impact on Crop Yield and Sustainability

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ABSTRACT

Effective nutrient management is central to agricultural productivity, yet only 30–50% of applied nitrogen is taken up by crops. Coated (controlled-release) fertilizers were developed to close this gap, but their release rate depends strongly on soil temperature and moisture, so a product rated for 90 days in the laboratory behaves very differently in the field, and current recommendation systems ignore these release kinetics. This paper presents an AI-driven dual-module system that links the two. Module 1 predicts the field release rate and cumulative release of polymer-coated urea from soil temperature, soil moisture, rainfall, soil pH and coating thickness. Module 2 combines this prediction with crop type, growth stage, initial N-P-K levels and soil pH to recommend the fertilizer type, application amount and timing. Because no public dataset records coated-fertilizer release under field weather, 2,200 real field records from the Kaggle Crop Recommendation dataset were expanded into 11,000 scenarios whose release was simulated from published kinetics. Implemented in Python with scikit-learn and XGBoost, Module 1 reached a test R^2 of 0.981 for release rate and 0.987 for cumulative release. Module 2 selected the fertilizer type with 85.7% accuracy, against a 88.4% ceiling set by label noise, and predicted amount and timing with R^2 of 0.992 and 0.976. For coated urea the system reduced application-timing error by 82% compared with the manufacturer's static rating, synchronizing nutrient supply with crop demand to improve sustainability and yield.

1. Introduction

The growing global population makes food security a pressing concern and calls for sustainable intensification of agriculture. Chemical fertilizers are one of the main ways to achieve high crop yields, but they are used very inefficiently: only an estimated 30–50% of applied nutrients are taken up by the crop. The rest is lost, which reduces farm profitability and causes serious environmental pollution, such as the eutrophication of water bodies. Advances in materials science led to coated fertilizers, also known as controlled-release fertilizers, which release nutrients gradually as the crop takes them up [2, 4]. In practice, however, the release pattern rarely matches the laboratory rating, because release depends strongly on soil temperature and moisture. A coated urea rated for 90 days may release its nitrogen in about 50 days in a hot, wet season and take 120 days or more in a cool, dry one.

This mismatch is a major gap for precision agriculture. Farmers have no tool to estimate how a coated fertilizer will actually behave under the conditions of their own field, and most fertilizer recommendation systems do not consider release kinetics at all. We address this gap with an intelligent dual-module system that first forecasts how a coated fertilizer will release its nutrients and then uses that forecast to produce a synchronized recommendation. The contributions of this work are:

- a dataset that couples 2,200 real soil and climate records with literature-based release kinetics;
- a release-prediction module (Module 1) that replaces the fixed label with a condition-specific estimate;
- a recommendation engine (Module 2) that consumes this estimate, including an engineered synchrony index; and
- an evaluation against the existing static practice and an ablation without Module 1.

2. Literature Review

Recent literature (2023–2025) from IEEE, Elsevier and Springer falls into three closely related research streams. The first applies machine learning to nutrient recommendation: models such as Random Forest, Support Vector Machines and XGBoost have been used to predict crop yield and to compute N-P-K rates for a given crop and soil [1, 11, 12]. These systems improve on current practice, but they assume that applied nutrients become available on a fixed schedule. The second stream uses IoT and sensor technology to capture field data such as soil moisture, temperature and electrical conductivity at high resolution [9, 11]. This is essential groundwork for any dynamic model, yet most studies stop at data collection and visualization rather than feeding the data into an agronomic model.

The third stream models nutrient release from slow- and controlled-release coated fertilizers. This research is usually carried out in the laboratory, where kinetic models are derived from diffusion and degradation laws [2, 3, 15]. These models describe release well under controlled conditions but do not account for how conditions change in the field, and recent reviews identify the lack of lab-to-field integration as an open research question [5, 14]. No integrated system currently uses real environmental data to predict how release will unfold in a specific field and then feeds that prediction into a recommendation engine [6, 10]. This paper proposes and evaluates an architecture that fills this gap.

3. Advantages or Merits

Combining AI with coated fertilizers offers the following benefits:

- Higher nutrient use efficiency: matching nutrient release to plant uptake means more of the applied fertilizer reaches the crop and less is left behind.
- Higher crop yield and quality: crops that receive nutrients at the right growth stage are more stress-resistant and give higher, more consistent yields.
- Lower environmental impact: reducing nutrient losses minimizes runoff and leaching of nitrogen (N) and phosphorus (P) into groundwater and surface water.
- Financial savings for farmers: yield targets can be met with less fertilizer, lowering material costs as well as the labour and fuel costs of repeated applications.

4. Disadvantages of Current Approaches

The proposed system is designed to overcome the following drawbacks of today's fertilizer management:

- **Static recommendations:** advice is based on an annual or biannual soil test and cannot adjust in season; a single heavy rainstorm can undo the year's fertilizer plan.
- **No account of environmental variability:** the effects of temperature and rainfall on nutrient release and loss change from season to season and from field to field.
- **Wrong-time fertilization:** nutrients are supplied too early or too late relative to crop demand, so more fertilizer is needed to reach the same yield.
- **High cost of inefficiency:** over-applying fertilizer as insurance against expected losses is economically wasteful and environmentally harmful.

5. Existing System

Farmers and agronomists mainly rely on two approaches today, and neither works well:

- **Soil-test-based recommendation:** a soil sample is sent to a laboratory, which returns a single static application rate. This approach is simple but ignores in-season dynamics and treats all fertilizers as if they released nutrients at the same rate.
- **Manufacturer's static rating:** a coated fertilizer is labelled with a fixed release period such as "90 days", measured in the laboratory at 25 °C. In the field, soil temperatures often exceed 35–40 °C, so the label does not describe real behaviour, and no decision support is provided for actual field conditions.

6. Tools Used

The prototype was built with standard open-source tools:

- **Python 3.9:** the programming language used for data processing and model implementation.
- **pandas 2.3 and NumPy 2.0:** data manipulation in DataFrames, numerical operations and generation of the simulated release data.
- **scikit-learn 1.6:** Linear and Polynomial Regression, Decision Tree, Random Forest, pipelines, cross-validation and evaluation metrics [18].
- **XGBoost 2.1:** gradient-boosted trees for the Module 2 recommendation engine [19].
- **Matplotlib 3.9:** visualization of data and results; **joblib** for saving trained models; **pytest** for 13 automated tests of the physics, the datasets and the integrated system.

7. Dataset and Data Preparation

7.1 Data sources

Two real datasets provide the soil and climate side of the problem. The Kaggle Crop Recommendation dataset [16] contains 2,200 field records (100 for each of 22 crops) with soil N, P and K levels, soil pH, air temperature, relative humidity and rainfall. A crop-wise N-P-K requirement table from the open-source Harvestify project [17] gives the agronomic target for each crop. No public dataset records the release of coated fertilizers under field weather, so release behaviour was simulated from relationships published in the literature, as described below.

7.2 Simulation of nutrient release

Each real record was expanded into five season scenarios. Soil temperature was derived as air temperature plus 1–6 °C (bare soil runs warmer than air) and clipped to 8–48 °C; volumetric soil moisture was derived from humidity and rainfall and clipped to 5–50%. Coating thickness L was drawn from 20–100 µm and days since application from 0–150. The manufacturer-rated duration (days to 80% release at 25 °C) grows with thickness:

$$D_{25} = 1.3\,L + 5.5 \quad (1)$$

so a 65 µm coat gives the standard 90-day product. Field conditions scale the release rate by a factor

$$k = Q_{10}^{(T-25)/10} \cdot f_M(\theta) \cdot f_R(R) \cdot f_{pH} \quad (2)$$

where $Q_{10} = 1.8$ reflects the finding that a 10 °C rise roughly halves release time [15], f_M lowers release in dry soil, f_R raises it slightly under heavy rainfall, and $f_{pH} = 1 + 0.04|\mathrm{pH} - 6.5|$ adds a weak pH effect. The actual field duration, mean release rate and cumulative release then

follow as

$$D = D_{25} / k, \quad r = 80 / D \qquad (3)$$

$$F(t) = 100\,[1 - \exp(-(t/\tau)^{\beta})], \quad \tau = D / (\ln 5)^{1/\beta} \qquad (4)$$

with shape β = 1.8, giving the S-shaped lag, release and plateau phases typical of polymer-coated urea and exactly 80% release at day D. To keep the learning problem realistic, 6% granule-batch variability, 3% measurement noise on the rate and ±1.5% noise on cumulative release were added. Fig. 2 shows the resulting release curves of a single 90-day product under three climates, and Fig. 3 shows that across the dataset this product actually lasts 40–126 days (5th–95th percentile, median 70 days), reproducing the 50–120-day field swing that motivates this work.

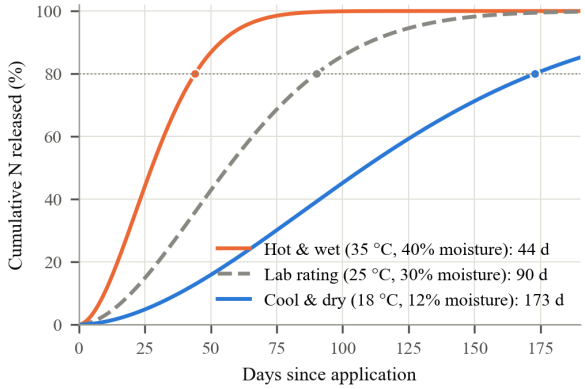


Fig. 2: Cumulative release of the same "90-day" polymer-coated urea under three field climates.

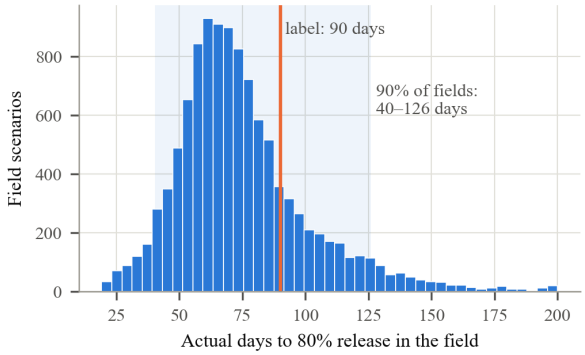


Fig. 3: Simulated field release duration of the "90-day" product across 11,000 field scenarios.

Recommendation labels for Module 2 were produced by agronomic rules. The most limiting nutrient (largest relative deficit against the crop target) decides the product: DAP for phosphorus, MOP for potassium, and NPK 19-19-19 when nutrients are co-limiting or no deficit is clear. For nitrogen, uncoated urea is chosen late in the season; otherwise the polymer-coated urea grade (PCU-60, PCU-90 or PCU-120) whose *field* duration best matches the crop's remaining demand window is chosen. The amount is the nutrient need divided by the product's nutrient content and a use efficiency that falls when release and demand are out of step; the timing is the number of days until the next application. Three percent of labels were replaced at random and decision thresholds were jittered to represent disagreement between agronomists.

7.3 Preprocessing and feature engineering

- **Missing values and duplicates:** none were found in the real data or in either generated dataset, so no imputation or de-duplication was required.
- **Outliers:** the IQR rule flags high P, K and rainfall values (for example, apple and grape soils with K ≈ 200). These are genuine crop-specific values and were kept; derived soil values were clipped to physical ranges.
- **Encoding and scaling:** crop type (22), growth stage (4) and fertilizer type (7) were one-hot encoded, and fertilizer classes were label-encoded for XGBoost. Features were standardized for the linear models; tree models used raw values.
- **Feature engineering:** a synchrony index, the predicted release duration divided by the days of crop demand remaining, was computed from Module 1's output and passed to Module 2.

Table 1: Datasets used in this study

Dataset	Source	Size	Content
Crop Recommendation	Real (Kaggle)	2,200 × 8	N, P, K, pH, temperature, humidity, rainfall; 22 crops
Crop N-P-K requirements	Real (Harvestify)	22 × 6	Target N, P, K per crop
Nutrient release	Generated (Module 1)	11,000 × 11	6 inputs → release rate, cumulative release
Recommendation	Generated (Module 2)	11,000 × 14	Crop, stage, N-P-K, pH, release → type, amount, timing

8. Proposed System

We propose an intelligent, dual-module AI-driven nutrient management system. The architecture first models the physical behaviour of the fertilizer and then uses that insight to drive a recommendation engine, as shown in Fig. 1.

Module 1: Nutrient Release Prediction. This module acts as a digital twin of the coated fertilizer. Its inputs are soil temperature, soil moisture and rainfall (environmental data), coating thickness (fertilizer data) and soil pH (soil data). It outputs the predicted release rate in percent per day and, given the days since application, the cumulative percentage released.

Module 2: Fertilizer Recommendation Engine. This module acts as an AI agronomist. Its inputs are crop type and growth stage, initial soil N, P and K levels and soil pH, together with the release rate predicted by Module 1 for a standard 90-day coated urea and the synchrony index derived from it. It works in two steps: it first classifies the fertilizer type among seven products (uncoated urea, PCU-60, PCU-90, PCU-120, DAP, MOP and NPK 19-19-19), and then predicts the application amount (kg/ha) and the number of days until the next application for that product. Predicting the product first matters

because the dose depends on the product: an NPK blend needs more than twice the mass of urea for the same nitrogen.

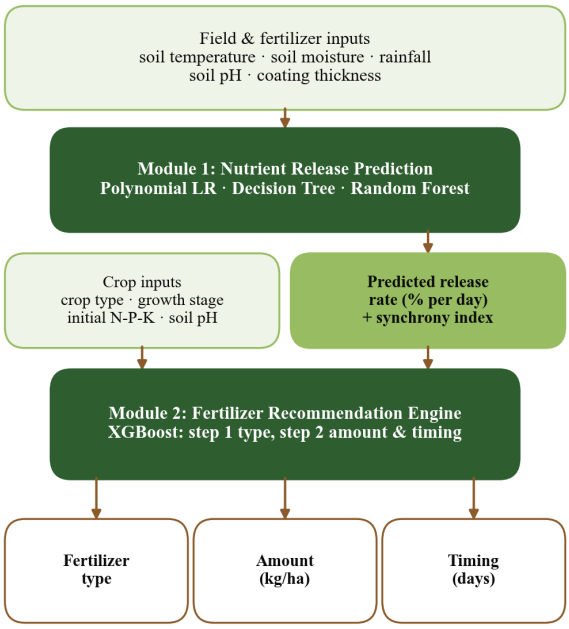


Fig. 1: System architecture of the proposed dual-module model.

Feeding Module 1's output into Module 2 is the key innovation: the recommendation is aware of how the fertilizer will actually behave in the field, rather than how it behaves in the laboratory.

9. Algorithms Used

Linear and Polynomial Regression (Module 1) provide an interpretable baseline for a physical relationship; degree-3 polynomial terms allow the linear model to follow the exponential temperature effect. **Decision Tree Regressor** (Module 1, and the Module 2 baseline) captures non-linear interactions, for example that high temperature combined with moist soil implies fast release, and provides feature importance. **Random Forest** averages many decision trees, which reduces variance and the risk of overfitting; it is used as a benchmark in Module 1 and as a candidate in Module 2. **XGBoost** builds trees sequentially by gradient boosting and handles mixed categorical and numerical inputs with complex interactions, which suits the recommendation task. Classifiers predict the fertilizer type; regressors predict the amount and timing.

All models were trained on an 80/20 train–test split (stratified by fertilizer type for Module 2). The best model for each target was selected by mean 5-fold cross-validation score on the training set, and the held-out 20% test set was used only once for the final evaluation. Regression was evaluated with MSE, RMSE, MAE and R^2; classification with accuracy, macro-F1 and the confusion matrix.

Table 2: Model hyperparameters

Model	Module	Key hyperparameters
Linear Regression	1	StandardScaler + ordinary least squares
Polynomial LR	1	Degree-3 polynomial features + OLS
Decision Tree	1, 2	M1: grid search over max_depth {6, 10, 14, None} and min_samples_leaf {1, 5, 10}; M2: depth 14, leaf 3
Random Forest	1, 2	300 trees; min_samples_leaf 1 (classifier) / 2 (regressor)
XGBoost	2	Classifier: 400 trees, depth 6, learning rate 0.08, subsample 0.9; regressors: 500 trees, depth 6, learning rate 0.06

10. Results and Discussion

10.1 Module 1: nutrient release prediction

Table 3 and Fig. 4 compare the four Module 1 models on the 2,200-row test set. Plain Linear Regression underfits (R^2 0.811) because temperature affects release exponentially. Adding degree-3 polynomial terms raises R^2 to 0.981, and this model was selected for the release rate (CV R^2 0.979). For cumulative release, which also depends on elapsed time, the Random Forest was selected with test R^2 0.987 (RMSE 3.96 percentage points). Decision-tree feature importance confirms the physics: coating thickness (57%) and soil temperature (40%) explain almost all of the release-rate variation, while days since application dominates cumulative release (76%). Fig. 5 shows close agreement between predicted and actual values.

Table 3: Module 1 test-set performance

Model	Rate R^2	Rate RMSE	Cum. R^2	Cum. RMSE
Linear Regression	0.811	0.396	0.809	15.04
Polynomial LR (deg 3)	0.981	0.124	0.960	6.88
Decision Tree	0.960	0.182	0.966	6.38
Random Forest	0.977	0.138	0.987	3.96

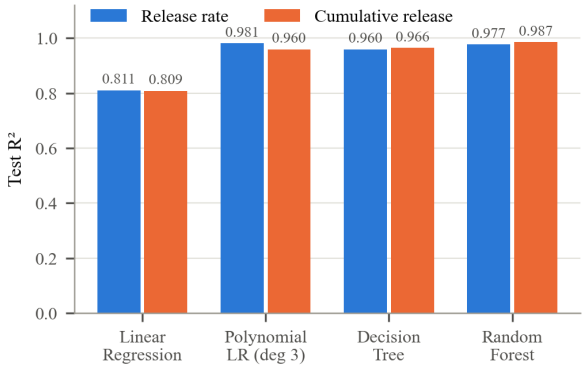


Fig. 4: Module 1 test R^2 by model for release rate and cumulative release.

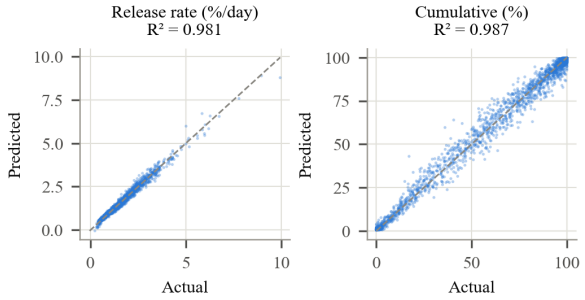


Fig. 5: Actual vs predicted values of the selected Module 1 models on the test set.

10.2 Module 2: fertilizer recommendation

Table 4 summarizes Module 2. XGBoost achieved the highest 5-fold CV score for all three targets (type CV accuracy 85.5%) and was selected as the final model. Its test accuracy of 85.7% should be read against the 88.4% obtained by the exact labelling rule itself, which is the maximum achievable given the injected label noise. The confusion matrix (Fig. 6) shows that most errors fall between neighbouring coat grades or involve the general NPK product. Given the recommended product, amount and timing reach R^2 0.992 and 0.976. When the product itself is predicted, end-to-end R^2 is 0.76 for amount and 0.83 for timing, since a wrong product implies a different dose.

Table 4: Module 2 test-set performance

Model	Type acc.	Macro-F1	Coated acc.	Amount R^2	Timing R^2
Decision Tree	80.4%	0.786	75.3%	0.987	0.961
Random Forest	83.0%	0.811	78.5%	0.992	0.973
XGBoost	85.7%	0.839	80.4%	0.992	0.976
XGBoost without Module 1	83.6%	0.789	71.0%	0.991	0.904

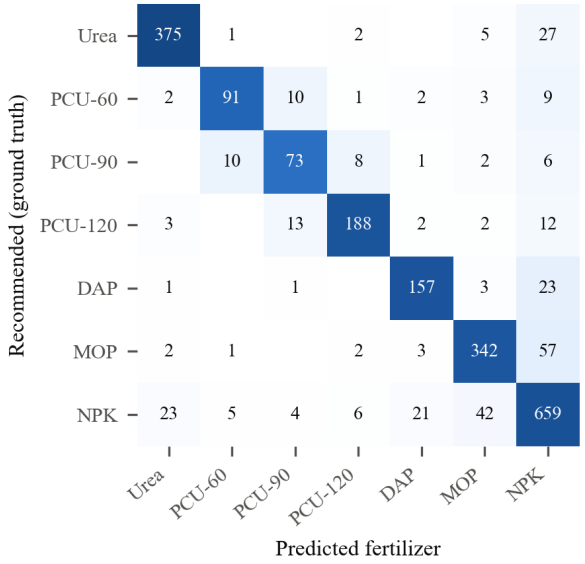


Fig. 6: Confusion matrix of the XGBoost fertilizer-type classifier on the test set.

10.3 Comparison with the existing static system

To test the central claim, the existing static approach, an ML recommender without Module 1, and the full dual-module system were given the same recommended product and compared on the test set (Table 5, Fig. 7). The static approach uses the manufacturer's rating, i.e. assumes laboratory conditions at 25 °C. For coated urea it mistimes the next application by 22.4 days on average; the proposed system reduces this to 4.0 days, an 82% reduction. The ablation confirms that the gain comes from Module 1: without the predicted release rate, coated-grade accuracy falls from 80.4% to 71.0% and timing R^2 from 0.976 to 0.904.

Table 5: Mean absolute error of the three systems on the test set

System	Timing MAE, all (days)	Timing MAE, coated urea (days)	Amount MAE (kg/ha)
Existing static system	9.2	22.4	12.6
ML without Module 1	4.4	8.4	9.0
Proposed dual-module system	2.4	4.0	8.4

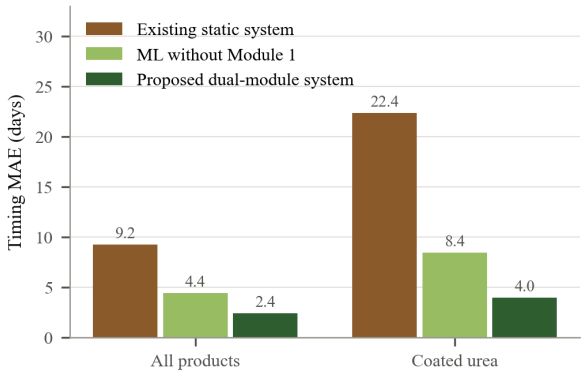


Fig. 7: Application-timing error of the static system, ML without Module 1, and the proposed system.

10.4 Integrated pipeline on user inputs

The integrated pipeline was tested through a command-line application that accepts user inputs, checks them against the training range and returns a recommendation. Table 6 shows two representative fields. In the hot, wet rice field, Module 1 predicts that the standard 90-day product would be exhausted in about 41 days, so Module 2 recommends the thickest coat. In the cool, dry maize field the same product would last about 144 days, so the thinnest coat is recommended. The two-stage process executed correctly from raw inputs to an actionable, synchronized fertilizer plan.

Table 6: Sample system inputs and outputs

	Hot, wet rice field	Cool, dry maize field
Inputs		
Soil temperature / moisture	36 °C / 42%	18 °C / 14%
Rainfall / soil pH	240 mm / 6.2	45 mm / 6.8
Growth stage	Sowing	Sowing
Soil N-P-K (kg/ha)	62-45-38	62-48-22
Module 1 output		
Release rate of 90-day PCU	1.93% per day	0.56% per day
Actual release duration	≈ 41 days	≈ 144 days
Module 2 output		
Recommended type	PCU-120 (71%)	PCU-60 (94%)
Recommended amount	246 kg/ha	223 kg/ha
Next application in	46 days	92 days

11. Conclusion

This paper presented a dual-module AI system that addresses the inefficiency of coated-fertilizer use by reconnecting static laboratory ratings with the dynamic conditions of the field. A release-prediction model (Module 1) feeds a recommendation engine (Module 2), so that fertilizer type, amount and timing are chosen with knowledge of how the product will actually behave. Module 1 predicted field release with R^2 of 0.98–0.99; Module 2, built on XGBoost, recommended the product with 85.7% accuracy and sized the dose and timing with $R^2 \geq 0.97$. Compared with the manufacturer's static rating, the system reduced coated-urea timing error by 82%, and an ablation showed that this improvement depends on Module 1.

The main limitation is that release behaviour and recommendation labels were simulated from published relationships rather than measured, so the models learn these relationships rather than observed field outcomes. Future work will (i) calibrate Module 1 with laboratory incubation or lysimeter release measurements for several fertilizers, (ii) collect historical soil, weather, application and yield data from partner farms, (iii) connect live weather forecasts and IoT soil sensors, and (iv) extend the system to sulphur-coated urea and coated P and K products. With such data, this architecture can become a practical decision-support tool that increases yield, reduces farmers' costs and improves the environmental footprint of modern agriculture.

12. GitHub Link

https://github.com/Amit-kr04/Smart-fertilizer-project

13. References

1. T. S. Tanaka, G. B. M. Heuvelink, T. Mieno, and D. S. Bullock, "Can machine learning models provide accurate fertilizer application recommendations?," *Precision Agriculture*, vol. 25, no. 4, pp. 1839–1856, 2024.
2. I. Kassem, E. H. Ablouh, F. Z. El Bouchtaoui, M. Jaouahar, and M. El Achaby, "Polymer-coated slow/controlled-release granular fertilizers: Fundamentals and research trends," *Progress in Materials Science*, vol. 144, art. no. 101269, 2024.
3. S. Swain et al., "Proportional impact prediction model of coating material on polymer-coated fertilizers: Diffusion release from single granules," *Scientific Reports*, vol. 14, art. no. 53410, 2024.
4. A. P. Dovzhenko et al., "Recent progress in the development of encapsulated controlled-release fertilizers," *Heliyon*, vol. 10, art. no. e12667, 2024.
5. S. Goyil, "Controlled-release fertiliser: Recent developments and future prospects," *Journal of Controlled Release*, vol. 366, pp. 233–249, 2024.
6. X. Li, "Global trends and current advances in slow/controlled-release fertilizers: A meta-analysis and synthesis," *Computers and Electronics in Agriculture*, vol. 220, art. no. 108456, 2024.
7. R. I. Rubel, M. S. Alam, and T. Khan, "Economic assessment of biochar-based controlled-release fertilizers," *Environmental Science and Pollution Research*, vol. 32, pp. 20401–20417, 2025.
8. N. Singh et al., "Encapsulated slow-release fertilizers: A review," *Environmental Nanotechnology, Monitoring & Management*, vol. 25, art. no. 100876, 2024.
9. J. Yin, "Agroecosystem modeling and precision agriculture for fertilizer optimization," *Agricultural Systems*, vol. 227, art. no. 103501, 2025.
10. T. M. C. Maaz, "A review of research and innovation on novel fertilizers for efficient nutrient management in agriculture," *Nature Food*, vol. 6, no. 3, pp. 155–168, 2025.
11. M. Padhiary, "Enhancing precision agriculture: A comprehensive review of machine learning and artificial intelligence in farming operations," *Computers and Electronics in Agriculture*, vol. 234, art. no. 106283, 2024.
12. M. Wu et al., "Precision forecasting of fertilizer components using machine learning and physical soil parameters," *Soil & Tillage Research*, vol. 214, art. no. 105469, 2024.
13. H. S. Hameed, "A review of nano-enabled controlled-release fertilizers," *Journal of Agricultural Engineering*, vol. 12, pp. 45–63, 2024.
14. A. Ravindran, "Advancements and challenges in controlled-release fertiliser: Recent developments and future prospects," *Journal of Controlled Release*, vol. 380, art. no. 101345, 2025.
15. M. J. Brown, R. E. Luebs, and P. F. Pratt, "Effect of temperature and coating thickness on the release of urea from resin-coated granules," *Agronomy Journal*, vol. 58, pp. 175–178, 1966.
16. A. Ingle, "Crop Recommendation Dataset," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset
17. Harvestify, "Crop-wise fertilizer requirement data," GitHub repository. [Online]. Available: https://github.com/Gladiator07/Harvestify
18. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
19. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.